Week 1: Foushia Begam I

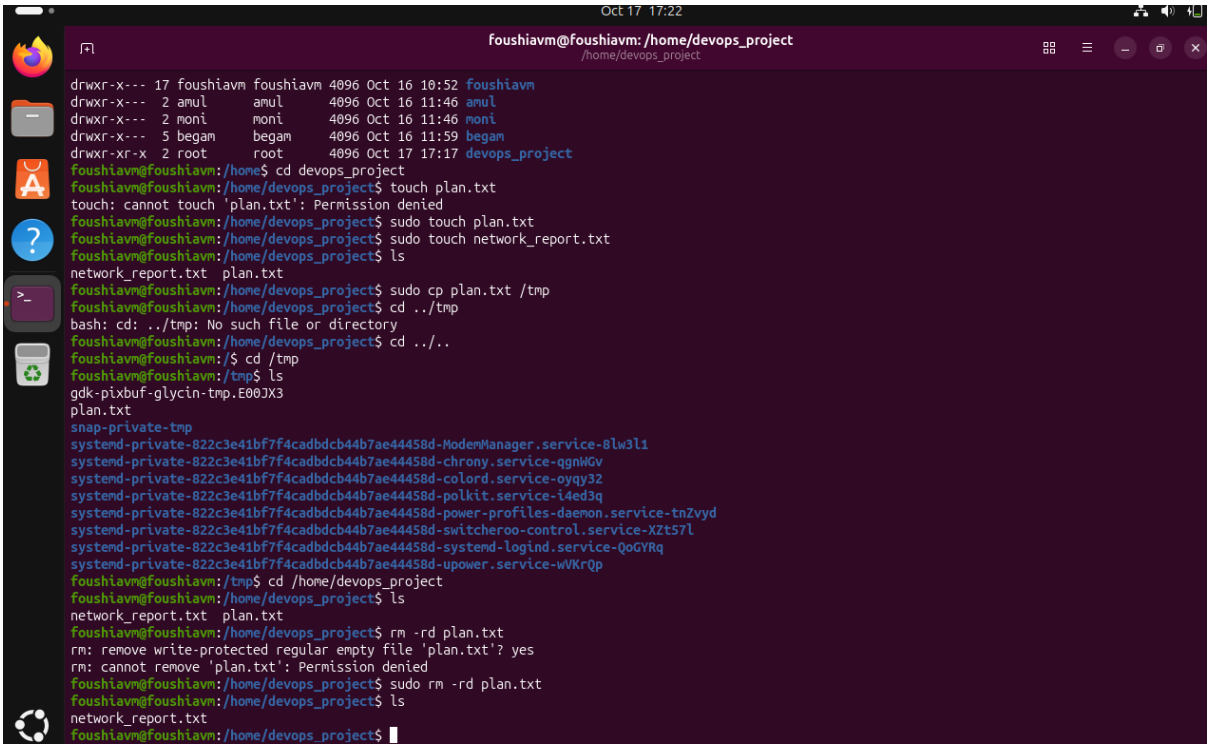
DVP-Week01-Assignment

Run and document output (screenshots or command logs) for the following:

1. Run a command to check the Linux distribution name and version of the operating system i nstalled on your machine.

Files & Directories:

- Create directory /home/devops_project
- Inside it, create 2 files: plan.txt and network_report.txt
- Copy plan.txt to /tmp and delete the original

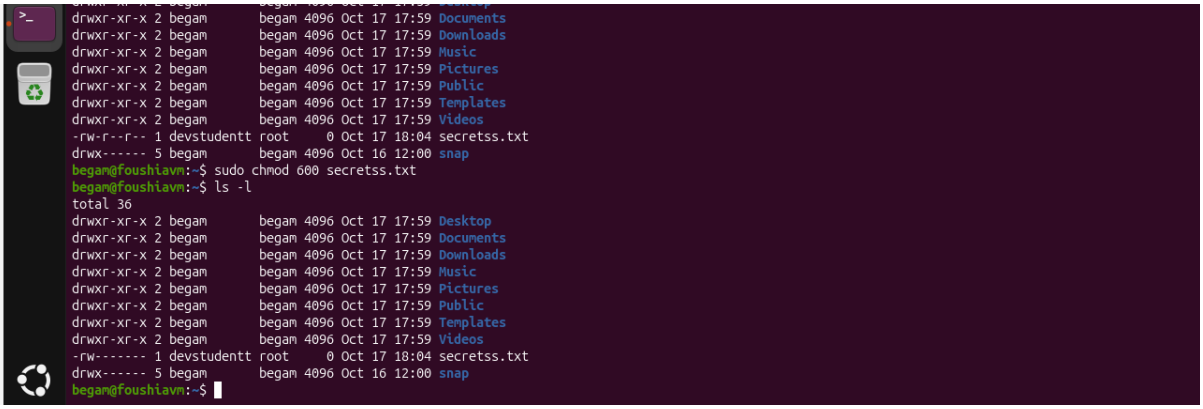


Users & Permissions:

- Create a user devstudent
- Create a group devops_team
- Add devstudent to devops_team
- Create a file secrets.txt, and:
- Change its owner to devstudent
- Set permissions so only owner can read/write

```
begam@foushiavm:~$ sudo adduser devstudentt
[sudo: authenticate] Password:
New password:
BAD PASSWORD: The password contains the user name in some form
Retype new password:
passwd: password updated successfully
Changing the user information for devstudentt
Enter the new value, or press ENTER for the default
        Full Name []: devstudentt
        Room Number []:
        Work Phone []:
        Home Phone []:
        Other []:

Is the information correct? [Y/n] begam@foushiavm:~$ y
y: command not found
begam@foushiavm:~$ sudo addgroup devops_teamm
begam@foushiavm:~$ sudo usermod -aG devops_teamm devstudentt
begam@foushiavm:~$ sudo usermod -aG sudo devstudentt
begam@foushiavm:~$ groups devstudentt
devstudentt : devstudentt sudo users devops_teamm
begam@foushiavm:~$ sudo touch secretss.txt
begam@foushiavm:~$ ls -l
total 36
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Desktop
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Documents
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Downloads
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Music
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Pictures
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Public
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Templates
drwxr-xr-x 2 begam begam 4096 Oct 17 17:59 Videos
-rw-r--r-- 1 root  root     0 Oct 17 18:04 secretss.txt
drwx------ 5 begam begam 4096 Oct 16 12:00 snap
begam@foushiavm:~$ sudo chown devstudentt secretss.txt
begam@foushiavm:~$ ls -l
total 36
drwxr-xr-x 2 begam       begam 4096 Oct 17 17:59 Desktop
drwxr-xr-x 2 begam       begam 4096 Oct 17 17:59 Documents
```



File Compression:

- Download a sample .log file[we can use **wget** command] from internet or create dummy one using echo
- Compress it using gzip and tar

```
begam@foushiavm:~$ echo "hello there!" > sample.log
begam@foushiavm:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  sample.log  secretss.txt  snap
begam@foushiavm:~$ gzip sample.log
begam@foushiavm:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  sample.log.gz  secretss.txt  snap
begam@foushiavm:~$ gunzip sample.log.gz
begam@foushiavm:~$ ls
Desktop  Documents  Downloads  Music  Pictures  Public  Templates  Videos  sample.log  secretss.txt  snap
begam@foushiavm:~$
```

Package Management:

- Install curl, htop, and net-tools using apt
- Show which and --version outputs

```
curl is already the newest version (8.18.0-1ubuntu2.8).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 73
begam@foushiavm:~$ sudo apt install htop
Installing:
  htop

Suggested packages:
  lm-sensors

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 73
  Download size: 177 kB
  Space needed: 451 kB / 22.7 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/main amd64 htop amd64 3.4.1-5build2 [177 kB]
Fetched 177 kB in 1s (128 kB/s)
Selecting previously unselected package htop.
(Reading database… 150915 files and directories currently installed.)
Preparing to unpack …/htop_3.4.1-5build2_amd64.deb…
Unpacking htop (3.4.1-5build2)…
Setting up htop (3.4.1-5build2)…
Processing triggers for desktop-file-utils (0.28-1build1)…
Processing triggers for hicolor-icon-theme (0.18-2build1)…
Processing triggers for gnome-menus (3.38.1-1ubuntu1)…
Processing triggers for man-db (2.13.1-1build1)…
begam@foushiavm:~$ sudo apt install net-tools
net-tools is already the newest version (2.10-2ubuntu1).
Summary:
  Upgrading: 0, Installing: 0, Removing: 0, Not Upgrading: 73
```

```
begam@foushiavm:~$ curl --version
curl 8.18.0 (x86_64-pc-linux-gnu) libcurl/8.18.0 OpenSSL/3.5.5 zlib/1.3.1 brotli/1.2.0 zstd/1.5.7 libidn2/2.3.8 libpsl/0.21.2 libssh2/1.11.1 nghttp2/
.68.0 librtmp/2.3 mit-krb5/1.22.1 OpenLDAP/2.6.10
Release-Date: 2026-01-07, security patched: 8.18.0-1ubuntu2.8
Protocols: dict file ftp ftps gopher gophers http https imap imaps ipfs ipns ldap ldaps mqtt pop3 pop3s rtmp rtsp scp sftp smb smbs smtp smtps telnet
tftp ws wss
Features: alt-svc AsynchDNS brotli GSS-API HSTS HTTP2 HTTPS-proxy IDN IPv6 Kerberos Largefile libz NTLM PSL SPNEGO SSL threadsafe TLS-SRP UnixSockets
zstd
begam@foushiavm:~$ htop --version
htop 3.4.1
begam@foushiavm:~$
```

Run and shortly explain outputs of the following commands:

- ip a or ifconfig

```
begam@foushiavm:~$ ip
Usage: ip [ OPTIONS ] OBJECT { COMMAND | help }
       ip [ -force ] -batch filename
where  OBJECT := { address | addrlabel | fou | help | ila | ioam | l2tp | link |
                   macsec | maddress | monitor | mptcp | mroute | mrule |
                   neighbor | neighbour | netconf | netns | nexthop | ntable |
                   ntbl | route | rule | sr | stats | tap | tcpmetrics |
                   token | tunnel | tuntap | vrf | xfrm }
       OPTIONS := { -V[ersion] | -s[tatistics] | -d[etails] | -r[esolve] |
                    -h[uman-readable] | -iec | -j[son] | -p[retty] |
                    -f[amily] { inet | inet6 | mpls | bridge | link } |
                    -4 | -6 | -M | -B | -0 |
                    -l[oops] { maximum-addr-flush-attempts } | -echo | -br[ief] |
                    -o[neline] | -t[imestamp] | -ts[hort] | -b[atch] [filename] |
                    -rc[vbuf] [size] | -n[etns] name | -N[umeric] | -a[ll] |
                    -c[olor]}
begam@foushiavm:~$ ifconfig
enp0s3: flags=4163<UP,BROADCAST,RUNNING,MULTICAST>  mtu 1500
        inet 10.0.2.15  netmask 255.255.255.0  broadcast 10.0.2.255
        inet6 fe80::a00:27ff:fe7b:9b58  prefixlen 64  scopeid 0x20<link>
        inet6 fd17:625c:f037:2:a00:27ff:fe7b:9b58  prefixlen 64  scopeid 0x0<global>
        ether 08:00:27:7b:9b:58  txqueuelen 1000  (Ethernet)
        RX packets 1061  bytes 731104 (731.1 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 984  bytes 98091 (98.0 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

lo: flags=73<UP,LOOPBACK,RUNNING>  mtu 65536
        inet 127.0.0.1  netmask 255.0.0.0
        inet6 ::1  prefixlen 128  scopeid 0x10<host>
        loop  txqueuelen 1000  (Local Loopback)
        RX packets 292  bytes 26233 (26.2 KB)
        RX errors 0  dropped 0  overruns 0  frame 0
        TX packets 292  bytes 26233 (26.2 KB)
        TX errors 0  dropped 0 overruns 0  carrier 0  collisions 0

begam@foushiavm:~$
```

- ping google.com -c 4

```
begam@foushiavm:~$ ping google.com -c 4
PING google.com (142.250.207.238) 56(84) bytes of data.
64 bytes from del12s11-in-f14.1e100.net (142.250.207.238): icmp_seq=1 ttl=64 time=48.9 ms
64 bytes from del12s11-in-f14.1e100.net (142.250.207.238): icmp_seq=2 ttl=64 time=48.1 ms
64 bytes from del12s11-in-f14.1e100.net (142.250.207.238): icmp_seq=3 ttl=64 time=48.1 ms
64 bytes from del12s11-in-f14.1e100.net (142.250.207.238): icmp_seq=4 ttl=64 time=51.3 ms

--- google.com ping statistics ---
4 packets transmitted, 4 received, 0% packet loss, time 4964ms
```

- traceroute google.com

```
begam@foushiavm:~$ sudo apt install traceroute
Installing:
  traceroute

Summary:
  Upgrading: 0, Installing: 1, Removing: 0, Not Upgrading: 73
  Download size: 54.4 kB
  Space needed: 167 kB / 22.7 GB available

Get:1 http://in.archive.ubuntu.com/ubuntu resolute/universe amd64 traceroute amd64 1:2.1.6-1build1 [54.4 kB]
Fetched 54.4 kB in 1s (38.8 kB/s)
Selecting previously unselected package traceroute.
(Reading database… 150925 files and directories currently installed.)
Preparing to unpack …/traceroute_1%3a2.1.6-1build1_amd64.deb…
Unpacking traceroute (1:2.1.6-1build1)…
Setting up traceroute (1:2.1.6-1build1)…
update-alternatives: using /usr/bin/traceroute.db to provide /usr/bin/traceroute (traceroute) in auto mode
update-alternatives: using /usr/bin/traceroute6.db to provide /usr/bin/traceroute6 (traceroute6) in auto mode
update-alternatives: using /usr/bin/lft.db to provide /usr/bin/lft (lft) in auto mode
update-alternatives: using /usr/bin/traceproto.db to provide /usr/bin/traceproto (traceproto) in auto mode
update-alternatives: using /usr/sbin/tcptraceroute.db to provide /usr/sbin/tcptraceroute (tcptraceroute) in auto mode
Processing triggers for man-db (2.13.1-1build1)…
begam@foushiavm:~$ traceroute google.com
traceroute to google.com (142.250.207.238), 30 hops max, 60 byte packets
 1  _gateway (10.0.2.2)  1.016 ms  0.852 ms  0.777 ms
 2  * * *
 3  * * *
 4  * * *
 5  * * *
 6  * * *
 7  *
```

- nslookup google.com

```
begam@foushiavm:~$ nslookup googl.com
Server:         127.0.0.53
Address:        127.0.0.53#53

Non-authoritative answer:
Name:   googl.com
Address: 192.178.211.106
Name:   googl.com
Address: 192.178.211.104
Name:   googl.com
Address: 192.178.211.147
Name:   googl.com
Address: 192.178.211.99
Name:   googl.com
Address: 192.178.211.103
Name:   googl.com
Address: 192.178.211.105
Name:   googl.com
Address: 2404:6800:4000:1025::6a
Name:   googl.com
Address: 2404:6800:4000:1025::63
Name:   googl.com
Address: 2404:6800:4000:1025::69
Name:   googl.com
Address: 2404:6800:4000:1025::68

begam@foushiavm:~$
```

- netstat -tuln

```
begam@foushiavm:~$ netstat -tulp
(Not all processes could be identified, non-owned process info
 will not be shown, you would have to be root to see it all.)
Active Internet connections (only servers)
Proto Recv-Q Send-Q Local Address           Foreign Address         State       PID/Program name
tcp        0      0 _localdnsstub:domain    0.0.0.0:*               LISTEN      -
tcp        0      0 localhost:ipp           0.0.0.0:*               LISTEN      -
tcp        0      0 _localdnsproxy:domain   0.0.0.0:*               LISTEN      -
tcp6       0      0 ip6-localhost:ipp       [::]:*                  LISTEN      -
udp        0      0 _localdnsproxy:domain   0.0.0.0:*                           -
udp        0      0 _localdnsstub:domain    0.0.0.0:*                           -
udp        0      0 0.0.0.0:mdns            0.0.0.0:*                           -
udp        0      0 localhost:323           0.0.0.0:*                           -
udp6       0      0 [::]:mdns               [::]:*                              -
udp6       0      0 ip6-localhost:323       [::]:*                              -
begam@foushiavm:~$
```

- curl ifconfig.me

```
begam@foushiavm:~$ curl ifconfig.me
42.104.122.193begam@foushiavm:~$
```